software.lucas-tvs.com: Launch Audit, Fix Status

Oct 30, 2026 · Status update on the 24-item launch audit · Changes are in the repo working tree, not yet deployed

Verdict

All four critical issues are fixed in code. 19 of the 24 items are fixed. Four are partly fixed, three of them because they need an account, key or ID only Lucas-TVS can supply. One is open because it needs the client to confirm the address. Every change builds cleanly (`npm run build`) and was checked on a local production build. The header, caching and 404 behaviour can only be confirmed after the next Vercel deploy.

Severity	Items	Fixed	Partly fixed	Open
Critical	4	4	0	0
High	6	4	2	0
Medium	8	7	1	0
Low	6	4	1	1
Total	**24**	**19**	**4**	**1**

Measured on the local production build

Check	Audit	Now
Hero video data on desktop	76 MB after 111 s; re-fetched on every rotation	~0.8 MB after 40 s; each clip fetched once
Five hero MP4s	18.6 MB	4.3 MB (0.5–1.2 MB each)
Hero on phones, Save-Data, reduced motion	Videos	Still image only, no video downloaded
Largest live-page PNGs (11 files)	~27 MB	~1 MB as WebP
Main JavaScript bundle	981 KB, one file	660 KB, with each inner page split into its own chunk
Routes with their own title, description, canonical and OG tags	1	44 (incl. /privacy-policy)
Smallest visible text on /contact	10 px	12 px

Lighthouse was not re-run: PageSpeed Insights tests the live URL, so scores update after the deploy.

Fix list

Critical: fix before any promotion

	Issue	Where	What was done	Status
1	**Lorem-ipsum certifications**	/quality	"Certified processes" section hidden behind `SHOW_CERTIFICATIONS = false`. No other placeholder copy on the page. **To re-enable:** supply the real certificate PDFs.	Fixed
2	**Placeholder Insights**	/insights and 9 articles	Hidden. Removed every link (404 page, homepage section), left Insights out of site search, set all Insights routes to `noindex` and left them out of the sitemap. Two flags bring it back: `SHOW_INSIGHTS` in routes.ts and `INSIGHTS_HIDDEN` in seo.ts.	Fixed
3	**Hero videos re-download forever**	Homepage hero	Each video now mounts once (no `src` swapping). The active slide preloads in full and the rest load metadata only. Playback pauses while the tab is hidden. Phones, Save-Data and reduced motion get the poster still. Clips re-encoded to 1280 px with no audio (18.6 → 4.3 MB). The slide dots now have a 24 px tap area.	Fixed
4	**No caching on fingerprinted files**	Sitewide	`/assets/*` is now served with `public, max-age=31536000, immutable` (vercel.json). Confirm the response header after deploy.	Fixed

High

	Issue	Where	What was done	Status
5	**Missing security headers**	Sitewide	Added X-Content-Type-Options, X-Frame-Options DENY, Referrer-Policy, Permissions-Policy and HSTS. CSP ships as `Content-Security-Policy-Report-Only`, covering the Apps Script, Maps and GA4 origins. **Next step:** after a week with no console reports, rename the key to `Content-Security-Policy`.	Fixed
6	**Enquiry form open to abuse**	/contact	Done: • Fields are capped and trimmed in the browser. • The repo's Apps Script copy rejects a filled honeypot, escapes values starting with = + - @, and caps each cell at 2,000 characters. **Still needed:** the script owner must paste in and redeploy the script. Moving submission to a Vercel function with Cloudflare Turnstile (and Resend email) needs Turnstile and Resend keys and a decision on who owns the data.	Partly fixed

7	**No analytics or conversion tracking**	Sitewide	GA4 runs under Consent Mode and the "Performance cookies" category of the new cookie preferences. It never loads under Do Not Track or Global Privacy Control. Page views are sent on every route change, and a `generate_lead` event fires on every successful enquiry. **Still needed:** set `GA_MEASUREMENT_ID` in `src/lib/analytics.ts`, mark `generate_lead` as a key event in GA4, and verify Search Console.	Partly fixed
8	**Crawlers and link previews see one page**	Sitewide	A build step writes static HTML for all 43 routes, each with its own title, description, canonical URL, Open Graph and Twitter tags. There is a 1200×630 OG image, and the homepage carries Organization JSON-LD. Tags also update on client-side navigation.	Fixed
9	**No robots.txt, sitemap or real 404**	Sitewide	Added robots.txt and a generated sitemap.xml (29 URLs; leaves out /search, the old homepage variants and Insights). Removed the catch-all rewrite, so unknown paths fall through to 404.html (noindex) with HTTP 404. /home3 now redirects permanently to /. **Confirm after deploy:** status codes and `/products?sector=…` links.	Fixed
10	**Oversized images**	Homepage sectors, sitewide	11 large PNGs (three sector images, Capabilities product shots, page backdrops) converted to WebP at about twice display size: ~27 MB → ~1 MB. Homepage images now set width and height, and below-the-fold images use `loading="lazy"`.	Fixed

Medium

	Issue	Where	What was done	Status
11	**Low colour contrast**	Sitewide	New text-only token `green-ink` (#1d7a32): • Lime eyebrows: 1.46 → 5.41:1. • Footer headings and "About Us": 2.28 → 5.41:1. • Small green labels: 4.15 → 5.41:1. • Mobile menu links: up to 18.4:1. • Hero sub-line: raised to 90% white with a soft shadow.	Fixed
12	**iPhone zooms into form fields**	/contact, homepage enquiry, /products search	All inputs, selects and textareas are now 16 px on phones.	Fixed
13	**Touch targets too small**	Sitewide	Footer links, the bottom-row links, breadcrumbs, hero dots and focus-grid dots now have tap areas of at least 24 px.	Fixed

14	**Form compliance gaps**	/contact	Required fields are marked with an asterisk and set `required` and `aria-required`. A "By submitting you agree to our Privacy Policy" line sits beside submit. Fields have maxlength caps (name 100, company 120, email 254, phone 30, message 2,000). The homepage enquiry pop-up got the same caps and consent line.	Fixed
15	**One 981 KB JavaScript bundle**	Sitewide	Every page except the homepage now loads with React.lazy, which cut the main bundle to 660 KB. Further gains would mean deferring the homepage's animation libraries (GSAP, Framer Motion).	Partly fixed
16	**Hidden pages**	Navigation	"Products & Platforms" added under Capabilities in the inner-page mobile menu. Both footers link to Products, Quality and Sustainability. Both headers have a search icon. /products lists all 15 products and has its own H1, "Products & Platforms." The desktop header still has no direct Products link.	Fixed
17	**Menu skips Electrification & Powertrain**	Menu, homepage	The homepage Electrification card now links to `/technologies#electrification`, so all five focus areas link to Technologies. The menu links to the Technologies page and doesn't list focus areas separately.	Fixed
18	**Broken definition lists**	Footer contact card, /contact	Each row is now a valid `<dl>` holding a `<dt>` and a `<dd>`.	Fixed

Low

	Issue	Where	What was done	Status
19	**Missing app icons**	Site root	Added favicon.ico, apple-touch-icon.png, 192 and 512 px icons, site.webmanifest and a theme-color meta tag.	Fixed
20	**Tiny text**	Footer, labels	All text is at least 12 px, including the footer Privacy Policy link (measured on /contact).	Fixed
21	**No llms.txt or security.txt**	Site root	Added both. security.txt points to the contact page; switch it to a security mailbox if one exists.	Fixed
22	**Footer address names INEL R&D Tech Center**	Footer	Unchanged. Lucas-TVS needs to confirm the correct address for this division.	Open
23	**Cookie banner covers a third of a phone**	Mobile	The banner was replaced by the new category-based consent banner (Reject all / Manage / Accept all, with a preferences dialog). On a 390×844 screen it now takes 26%, down from about a third. A further compact phone layout is optional polish.	Partly fixed

24	**Sticky header hides anchored headings**	Anchored sections	Every element with an id now gets `scroll-margin-top: 104px`.	Fixed

Needs Lucas-TVS or the account owner

- **Real certificate PDFs**, to bring the /quality certifications back (item 1).
- **Real articles**, to bring Insights back (item 2).
- **Apps Script redeploy:** paste `scripts/google-sheets/enquiries.gs` into the live script and publish a new version. Also answer the audit's open question about which Google account owns the script and the sheet (item 6).
- **Turnstile and Resend keys**, plus an enquiries inbox Lucas-TVS owns, to move the form to a Vercel function (item 6).
- **GA4 measurement ID** (`GA_MEASUREMENT_ID` in analytics.ts) and **Search Console verification** (item 7).
- **Confirmation of the footer address** (item 22) and, optionally, a security contact mailbox (item 21).
- **Client review:** /products now shows all 15 products by default, and the homepage Electrification card links to /technologies instead of the external products page.

Verify after deploy

- `curl -I` a file under `/assets/` : it should return `Cache-Control: public, max-age=31536000, immutable`.
- `curl -I /made-up-page` : it should return HTTP 404. `/about` should return 200 with its own title, and `/about/` should redirect to `/about` .
- Check the security headers with securityheaders.com, then watch the console for CSP report-only warnings for a week.
- Re-run PageSpeed Insights (mobile and desktop), and paste a page link into LinkedIn or WhatsApp to check the preview card.

Method

Fixes were made in the repository and checked with `npm run build` and the TypeScript compiler. The production build was then served locally (`vite preview`) and tested in a browser:

- **Hero video:** network transfer and playback state at 1366 px and 640 px widths.
- **Static HTML:** per-route head tags in the prerendered pages.
- **Contact form:** font sizes, required flags, maxlength and the privacy link.
- **Pages:** no lorem-ipsum or Insights links on /, /quality, /products and /search; valid definition lists; minimum text size; footer link heights.
- **Console:** no console errors.

Not covered: Vercel-only behaviour (headers, caching, 404 status), Lighthouse scores, physical devices, and a live form submission.